

Andrey Zakharov

Backend Developer

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Backend Developer with 3+ years of hands-on project-based development experience. Builds API-driven services and backend-heavy applications in Python and Java, including data processing, external integrations, simulation engines, AI/media workflows and asynchronous messaging platforms. Experienced in taking projects from architecture and MVP implementation through testing, packaging and deployment-ready workflows. Open to working with Go, PHP and other backend ecosystems.

Experience

Independent Project Development: 2023 – Present.

COMPASS AI – Python Backend & Applied AI Engineer

2026 – Present

Tech: Python 3.11, FastAPI, Pydantic, Pandas, NumPy, REST APIs, Plane API, CSV/JSON/Parquet, ONNX Runtime, Docker

- Designed and implemented a modular FastAPI backend for task assignment, employee matching, bulk workflows, recommendation sessions, reporting and Plane project-management integration.
- Built validated CSV, JSON and Parquet ingestion/export, feature-preparation, training-session and inference workflows across 42,052 training rows and 8,924 test rows; added automated tests for matching, importers and end-to-end pipelines.
- Integrated Plane REST API and local Qwen/Ollama services with modular orchestration, explainability, fairness reporting and recommendation workflows; evaluated ranking with ROC-AUC 0.859, MRR 0.944 and P@1 0.893.

UrbanFlow AI – Python Backend & Simulation Engineer

2026

Tech: Python, FastAPI, Uvicorn, Pydantic, httpx, Pandas, SUMO, TraCI, OpenStreetMap, JSON/JSONL

- Designed a FastAPI backend coordinating OpenStreetMap import and normalization, SUMO scenario generation, simulation sessions, road-editor patches, training jobs, metrics and model artifacts.
- Integrated SUMO and TraCI to control traffic lights and simulation entities, including vehicles, pedestrians, public transport and dynamically applied road events.
- Implemented experiment tracking, model registry, JSON checkpoints and generated analytics for simulation runs with 300–500 vehicles and up to 2,241 controlled traffic lights.

Cartopia – Java Backend & Geospatial Software Engineer

2023 – 2025

Tech: Java 17, Forge, Gradle, Jackson, Gson, Java HttpServer, OpenStreetMap, Overpass/Nominatim, DEM/GeoTIFF, OpenLandMap, GMRT/OpenTopography, Open-Meteo

- Built a Java/Forge geospatial generation backend integrating OpenStreetMap, Overpass/Nominatim, DEM/GeoTIFF elevation, land-cover data, weather services and procedural world-generation components.
- Implemented a local Java HTTP server and endpoints /build, /player, /save-coords, /health and /realtime for place search, bounding-box selection, generation control, progress tracking and runtime interaction.
- Designed an extensible architecture with 80+ generator and storage classes, streaming NDJSON sidecars and terrain-grid storage; added validation, clipping, retries and cleanup for external geodata sources.

Face-Swap-Studio – Python Backend Engineer

2025

Tech: Python, FastAPI, Pydantic, OpenCV, NumPy, InsightFace, ONNX Runtime, file processing, macOS

- Built a FastAPI service for session lifecycle, uploads, face detection, visual mapping, image generation, batch processing and result archives.
- Designed a model registry and readiness layer for nine internal and external backends, separate runners, multi-face processing, per-face assignments, safe file handling and temporary-data cleanup.

Botyan – Async Python Backend Engineer

2023 – 2024

Tech: Python, asyncio, aiogram, discord.py, OpenRouter/OpenAI-compatible APIs, requests/httpx, JSON, Pillow, MoviePy, ffmpeg

- Built an asynchronous Python service running Telegram and Discord clients in parallel with shared conversation history, command handlers, access control, logging and message deduplication.
- Implemented JSON-backed persistence for messages, balances, daily rewards, slots and API-key limits, supporting context windows of up to 100 messages for chat and 1,000 for image captions; added Pillow, MoviePy and ffmpeg workflows for captions, demotivators and Telegram video-note effects.

Skills

Technical: Python 3.11, FastAPI, Pydantic, asyncio, Uvicorn, Java 17, Java HttpServer, JavaScript/TypeScript, Node.js/Express, REST APIs, Pandas, NumPy, CSV/JSON/Parquet, ONNX Runtime, OpenCV, aiogram, discord.py, Gradle, Docker Compose, Git. Open to working with Go, PHP and other backend languages.

Languages: English – C1 (Advanced); Russian – Native.

Education

SAINT PETERSBURG STATE UNIVERSITY OF AEROSPACE INSTRUMENTATION (SUAI) – Saint Petersburg, Russia

- Master's Degree, Management – Strategic Human Resource Management 2027
- Bachelor's Degree, Software Engineering – Software Systems Design 2025
- Professional Retraining Diploma – Web and Multimedia Applications Developer 2024
- Additional Professional Education: 250+ certificates and diplomas from 60+ universities, companies, educational platforms, institutes, and professional training programs worldwide, covering diverse academic and professional fields and disciplines.